Behavior of Perpetual Futures at Inception:

U.S. Kalshi Markets

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Abstract:

This year Kalshi became the first CFTC regulated venue in the United States to be allowed to offer perpetual futures, launching a variety of crypto currency contracts that went live in five waves over a three week window from June 3rd to June 24th, 2026. We will use this rollout to study perpetual future patterns at inception till August 13th, 2026; whereas international venues launched years ago without available data. Pooling the thirteen launches shows funding rates are systematically negative in the first two weeks after listing where the perpetual trades below its reference price and shorts pay longs, decaying to a deadband dominated near zero state within roughly three weeks. The decay is significant in a pooled regression (slope -0.0117 bps/day, $p<0.001$, $n=2539$) and survives clustering by market ($p=3.6x10^{-4}$), leave one market out re-estimation shows one market does not dominate results, and sign test shows the share of negative funding reading falls from 91.9% in the first two weeks to 55.4% after day 21. An exponential decay fit, motivated by Ackerer, Hugonnier and Jermann's (2025) constant hazard rate convergence theory, fits the data better than a straight line and implies a mispricing half-life of roughly 15 days. Verification checks rule out the results being an artifact of Kalshi's own pricing construction, and secondary hypotheses such as: already liquid coins converge faster, funding co-moves across markets, or liquidity formation predicts price discovery speed, do not find support in this sample.

Introduction:

Perpetual futures are the dominant derivative instrument in cryptocurrency markets, but until 2026 there was no venue offering them operating within U.S. federal derivatives regulation. In May 2026, Kalshi, a prediction market regulated by the CFTC, was granted permission to launch perpetual futures on a slate of cryptocurrencies, becoming the first market in the U.S. There were five waves of launches between June 3rd and 24th of 2026 starting with Bitcoin, Ethereum, Solana and XRP. The staggered rollout provides a unique opportunity to study these markets at inception, as they have not yet found their liquidity or natural clearing price, and cannot alternatively be studied on mature venues such as the prior work of He, Manela, Ross and von Wachter (2024) or venues' published inception data from 2016, of solely bitcoin in an obsolete market environment.

Contract Mechanics:

A perpetual future is a cash settled derivative that offers leveraged exposure to an underlying asset without an expiration date. A trader opens a long or short positions by posting margin, collateral equal to a fraction of the position's full value. Positions are marked to market continuously so as the contract's price moves, unrealized gains and

losses are credited or debited to the trader's margin account in real time. To protect the exchange, if a position's margin balances fall below a maintenance threshold, the exchange can force liquidation using the remaining margin and occasionally additional insurance to cover the loss. Since the contract never expires there is never delivery of the underlying asset.

The Funding Mechanism:

The funding rate will serve as the anchor of the contract, in the absence of an expiration date as in dated contracts. At each funding interval, every eight hours on Kalshi and industry standard cadence, the exchange computes a funding rate from the recent history of the gap between the contract's own price and its reference asset price. If the perpetual has been trading below its reference price, shorts pay longs and vice versa. The imbalance will dictate order flow, and the market will push the price back toward the reference. Kalshi collects revenue through a tiered trading fee schedule, market makers provide liquidity through bid ask spreads especially during mispricing, and arbitrageurs can opportunistically carry the underlying asset as empirically studied by He, Manela, Ross and von Wachter (2024) and formalized theoretically by Ackerer, Hugonnier, and Jermann's (2025).

Related Literature:

"Measuring Asset Values for Cash Settlement in Derivative Markets: Hedonic Repeated Measures Indices and Perpetual Futures." Robert J. Shiller (1993) proposed the theoretical groundwork for perpetual futures contacts as a solution to pricing assets with infrequently observed prices such as real estate, labor costs or commercial rents. The perpetual claim could settle against a periodic dividend or rent index that would provide continuous price discovery.

"Fundamentals of Perpetual Futures" He, Songrun, Asaf Manela, Omri Ross, and Victor von Wachter (2024) empirically study five major cryptocurrency perpetuals on Binance from January 2020 to March 2024. Their findings showed mean annualized deviation of the perpetual price from its no arbitrage benchmark is close to zero, but the absolute deviation is large 60-90% annualized, deviations across coins are correlated, and deviations shrink by roughly 11% per year over their sample as growing arbitrage capital competition joined the market.

"Perpetual Futures Pricing" Ackerer, Hugonnier, and Jermann (2025) formalize the no arbitrage pricing theory through risk neutral expectations of the future spot price sampled at random times governed by funding intensity. Also finding the intensity deviation decays exponentially and not linearly.

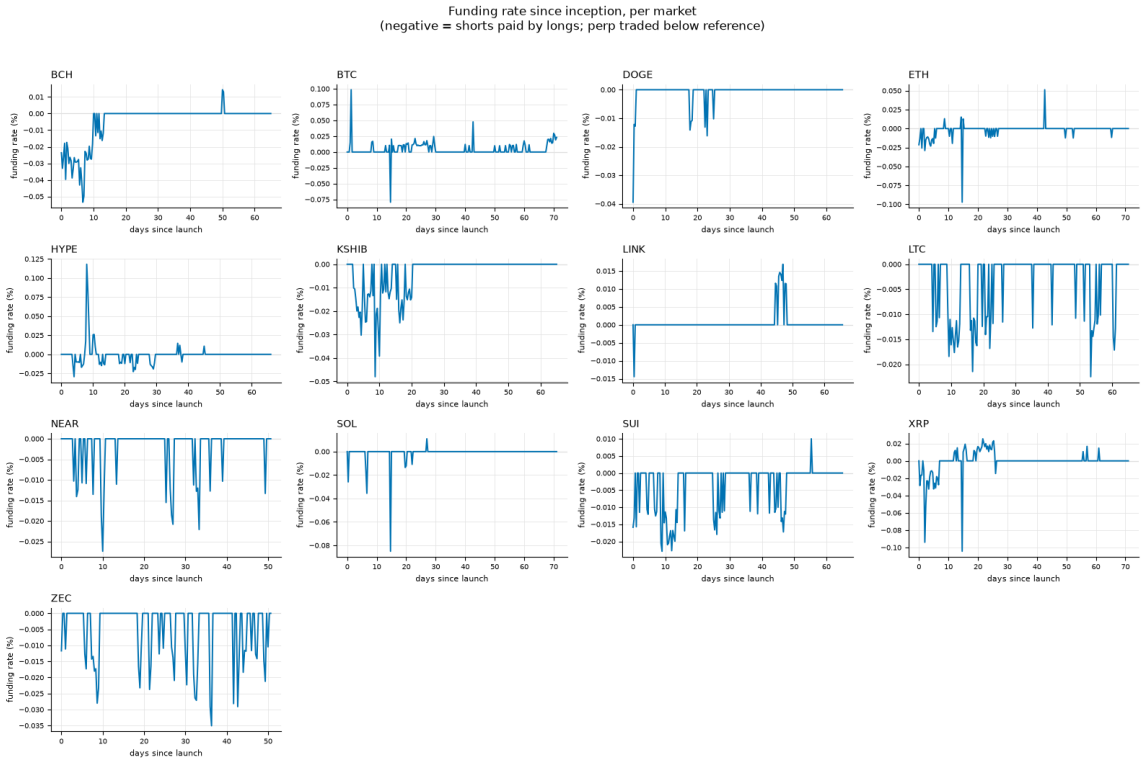
Data:

Collected data from Kalshi's perpetual market, OKX as an offshore perpetual candlestick reference (Binance and Bybit public endpoints refused connection from the U.S.), and CoinGecko as an hourly external spot reference for each coin and Kalshi does not have its own spot crypto market.

Kalshi's first trade of perpetual futures occurred on June 3rd, 2026, on BTC, ETH, SOL, and XRP reaching volume of over 480,000 within nine hours. Subsequent launches HYPE June 8th, BCH, DOGE, KSHIB, LTC, SUI and LINK June 9th, and NEAR and ZEC June 24th. DOT, HBAR, and XLM are listed with zero volume and are excluded from the analysis. Retrieved market snapshot of current price, bid/ask, open interest, contract size, daily candlestick open close high low, and funding rate marked price, timestamped every 8 hours. 2539 observations ending August 13th, 2026.

Results:

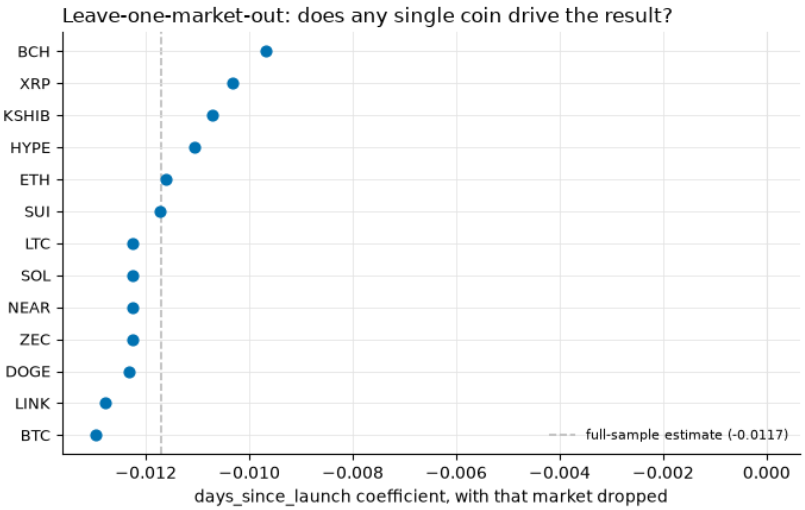
Days since launch is computed as trading time elapsed since each market's own first funding observation and all thirteen markets are aligned on the shared event time clock rather than on calendar date. Plotting funding rate against days since launch, aggregated and separately for each market, shows a consistent pattern of negative funding, where the perpetual trades below its reference price, and shorts pay longs, clustered in the first ten to fifteen days after launch and decaying toward flat after twenty one days.



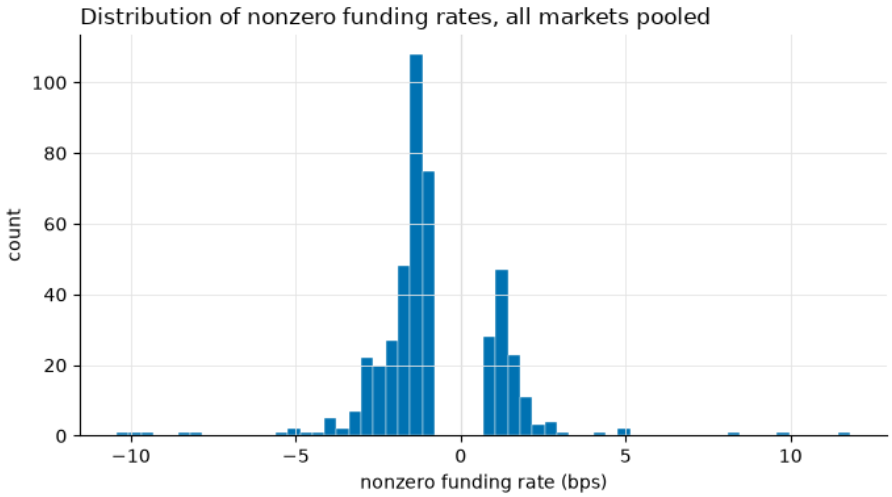
The findings are formalized with a pooled OLS regression of |funding rate| in basis points on days since launch, across thirteen markets with heteroskedasticity robust standard errors. The model predicts mispricing right at launch is about 0.70 bps shrinking by roughly 0.012 bps per day. HC p value of 9.3x10^-30 overstates confidence treating each reading independently when there are thirteen individually aligned markets.

Treating each market as independent units and clustering standard errors by market leaves the point estimate unchanged and is a more realistic p value of 3.6x10^-4.

Rerunning the regression thirteen times and dropping one market each trial leaves the coefficient negative and significant in all thirteen cases showing no single coin drives the results.

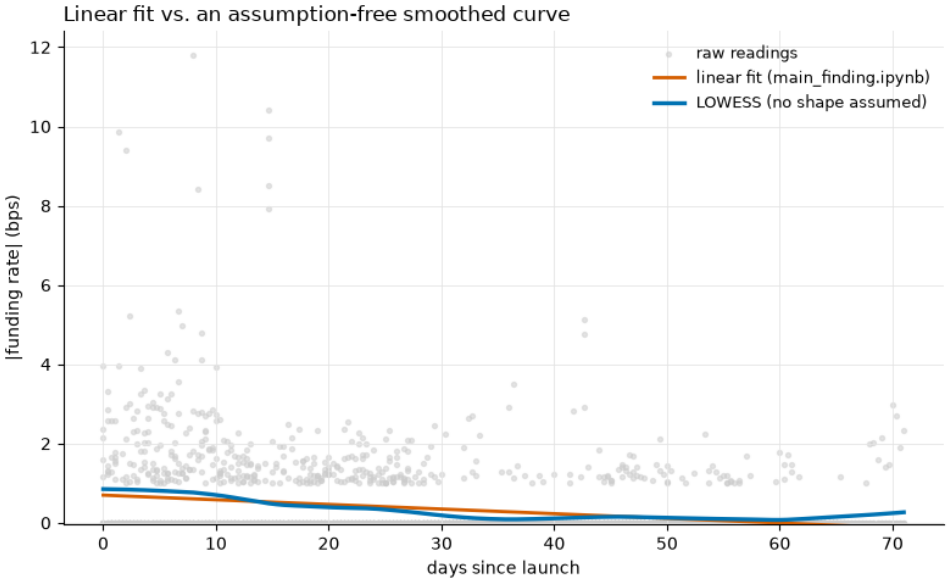


There is evidence of a deadband in the data. Only 17.6% of the readings are non-zero, clustering around a 1bps floor.

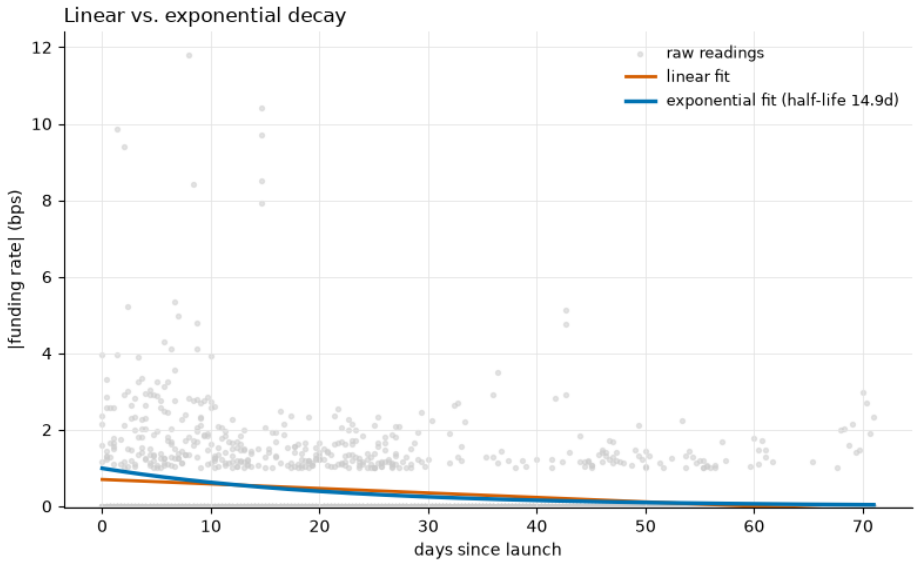


Of the nonzero readings, negative funding observations fall from 91.9% in days 0-14 to 55.4% after day 21 supported by a two proportion z test, z=8.18, p=2.8x10^-16.

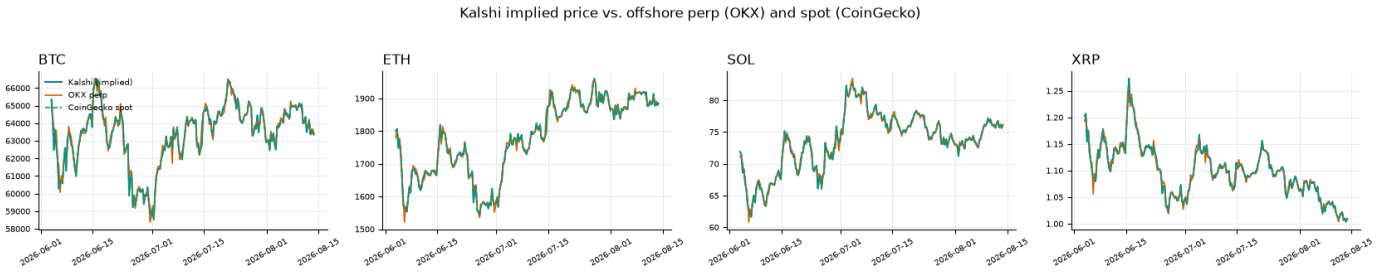
A locally weighted (LOWESS) tests traces a curve that plateaus over the market maturing, closer in shape to an exponential decay.



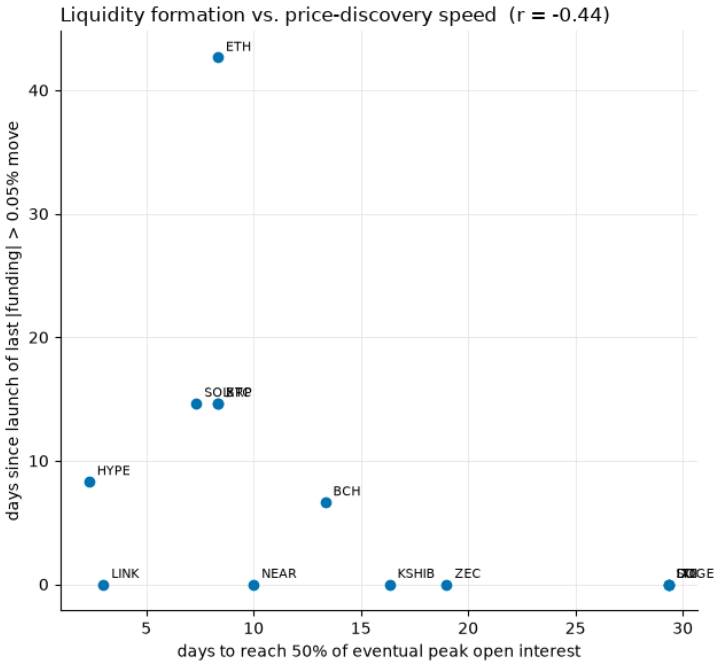
This motivates fitting the decay via nonlinear least squares. Resulting in a decay rate of b=0.0466 per day implying a mispricing half-life of 15 days. The exponential fit describes the data better than the linear one, pseudo R^2 0.084 vs linear R^2 0.067, consistent with Ackerer, Hugonnier and Jermann's (2025) theoretical prediction of exponential convergence.



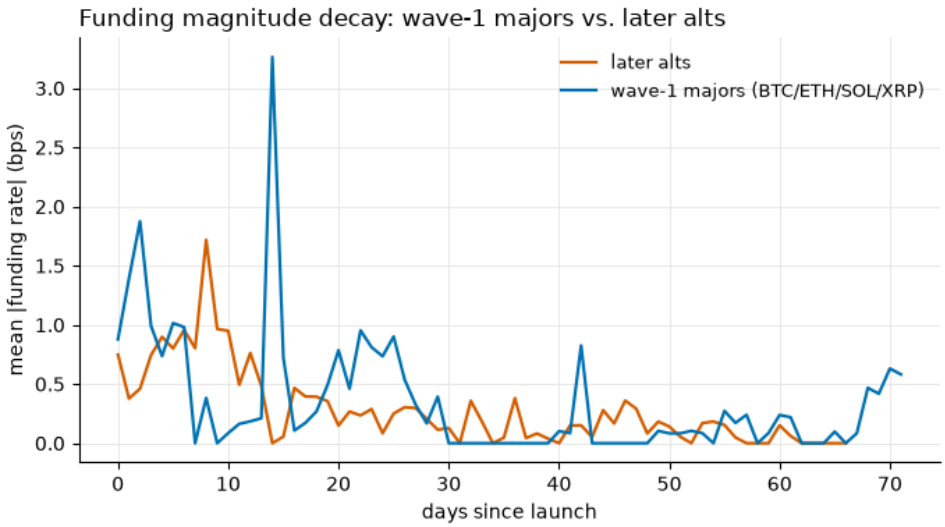
Verifying Kalshi's mark price to a per coin figure, by dividing the price by the contract size, and comparing it to OKX's perpetual price and CoinGecko's spot price to the nearest two hour timestamp shows Kalshi's markets track both external references closely over the full sample.



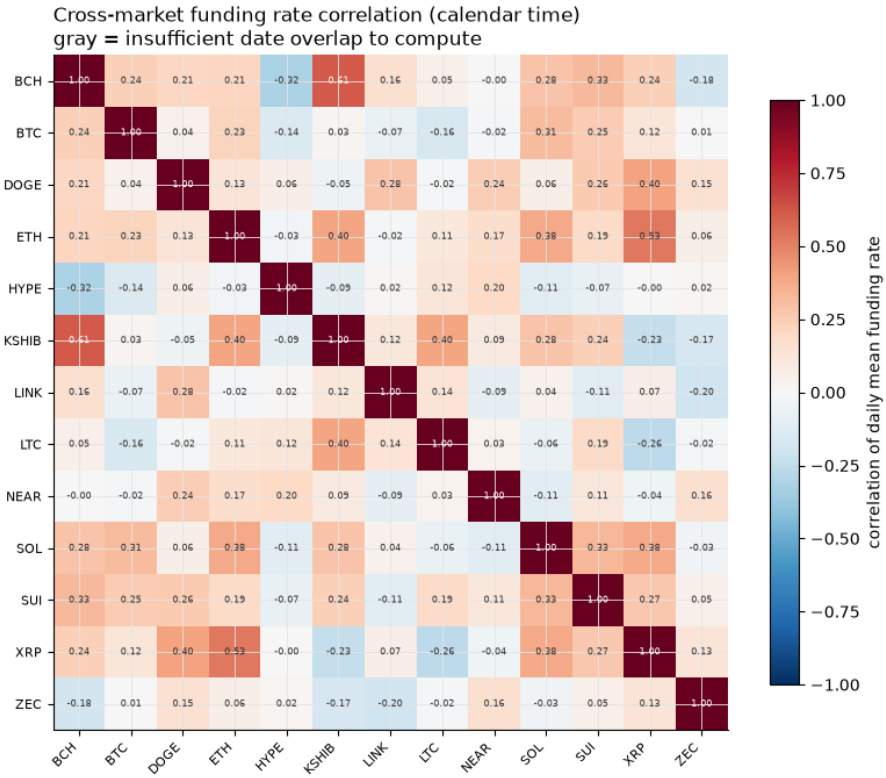
Checking if liquidity formation predicts price discovery speed, by correlating each market's days to half of eventual peak open interest against its days since last large funding move, surprisingly yields a negative correlation driven substantially by a single outlier so the result is mostly noise.



Further, adding a 'tier one' indicator for BTC, ETH, SOL, and XRP, the four markets with substantial preexisting liquidity, yields no significant difference from the remaining nine.



Additionally correlating daily mean funding rate across all thirteen markets yields a mean pairwise correlation of 0.097, median 0.082, substantially weaker than predicted in previous works, and indicating a lack of market makers and arbitrageurs.



Conclusion:

Kalshi's 2026 launch of the first CFTC regulated perpetual futures provides a unique vantage point on market inception of thirteen staggered, independent launches of well understood instruments by a regulated venue of a very liquid reference market. This paper

documents a robust, statistically significant pattern of negative funding in the first two weeks after listing, decaying approximately exponentially with an estimated half-life of 14.9 days. The pattern is consistent with a market where dedicated arbitrage and market making capacity has not yet arrived on day one but accumulates over roughly three weeks. The product ultimately reliably tracks the underlying, and uniquely offers: leverage, no expiration avoiding rollover costs and risks, and no custody burden allowing for shorting without first borrowing the underlying. The sample is small and underpowered, future work would continue to track these markets as well as new ones to check for consistent patterns. Currently Kalshi has a pending CFTC filing for gold, silver, copper and platinum perpetuals, as well as an equity index. CME group has filed suit against CFTC, claiming perpetual futures are truly more like swaps requiring stronger regulation.

References:

Ackerer, D., Hugonnier, J., and Jermann, U. (2025). “Perpetual Futures Pricing.” *Mathematical Finance*, 36(3), 481–499.

He, S., Manela, A., Ross, O., and von Wachter, V. (2024). “Fundamentals of Perpetual Futures.” Working paper, Washington University in St. Louis.

Shiller, R. J. (1993). “Measuring Asset Values for Cash Settlement in Derivative Markets: Hedonic Repeated Measures Indices and Perpetual Futures.” *Journal of Finance*, 48(3), 911–931.